Data imbalance means that the classes in your dataset are not equally represented — one class has way more samples than the others.

Example:

In a disease detection dataset:

- 95% samples = Healthy
- 5% samples = Diseased
  This is an imbalanced dataset.

Why it's a problem:

- Model learns to predict only the majority class.
- It gives high accuracy but poor real performance, especially for minority class.

Solutions:

- Oversampling minority class (e.g., SMOTE)
- Undersampling majority class
- Class weighting in models
- Anomaly detection methods (if minority class is rare/critical)

Techniques:
1. Oversampling

Increase samples of the minority class
🔹 Technique: SMOTE (Synthetic Minority Oversampling Technique)

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2. Undersampling

Reduce samples from the majority class
🔹 Technique: Random Undersampling

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3. Class Weighting

Give more importance to minority class during training
🔹 Used in: Logistic Regression, SVM, Neural Networks

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4. Ensemble Methods

Combine multiple models to balance learning

- 🔹 Technique: Balanced Random Forest, EasyEnsemble

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5. Anomaly Detection

Treat minority class as rare events or outliers

- 🔹 Used in: Fraud or disease detection

What is SMOTE?

SMOTE (Synthetic Minority Over-sampling Technique) is a technique to handle data imbalance by creating synthetic (fake but realistic) samples of the minority class, instead of just duplicating existing ones.

⚙️ How it works:

1. For each sample in the minority class, SMOTE finds its k nearest neighbors.
2. Then it creates a new sample along the line between the sample and one of its neighbors.
3. This helps to generalize the minority class area in feature space.

🔸 Types of SMOTE:

Type	Purpose	Use Case
Regular SMOTE	Generates synthetic data between minority class samples	General use
Borderline-SMOTE	Focuses on samples near class boundaries	When classes overlap
SMOTE-NC	Handles categorical + numerical features	Mixed data
KMeans-SMOTE	Uses clustering before oversampling	When data has subgroups in minority class
ADASYN (variant)	Focuses more on hard-to-learn examples	When some minority points are harder to classify